CDS Indices

Seminar Module

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Topics Covered

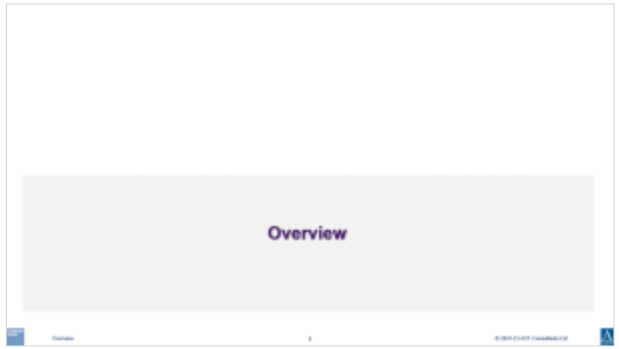
- Overview of Credit Derivative Indices
- CDX Indices
- iTraxx Indices
- Market Statistics
- Pricing an Index

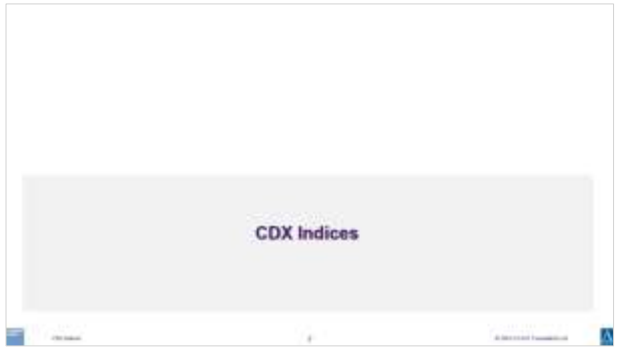
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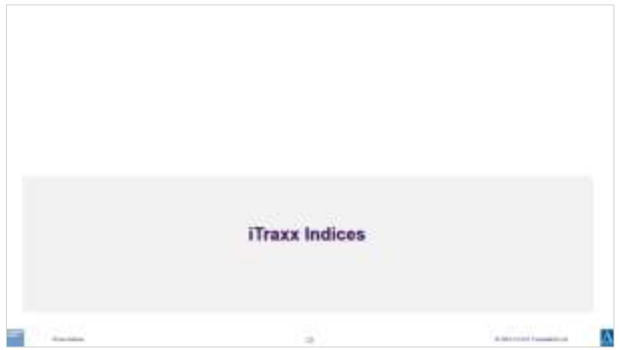
Key Point Summary

- CDS indices are linked to the prices of a basket of underlying single-name CDSs
- There are two global index families: CDX and iTraxx
- A new series of each liquid index is created every six months
- Index premiums are standardised and involve up-front payments – like those of single-name CDS
- If a credit event occurs, cash settlement is made on the pro-rata notional
- The majority of index trading now takes place on SEFs
- The fair index premium is the weighted average of the underlying single-name premiums

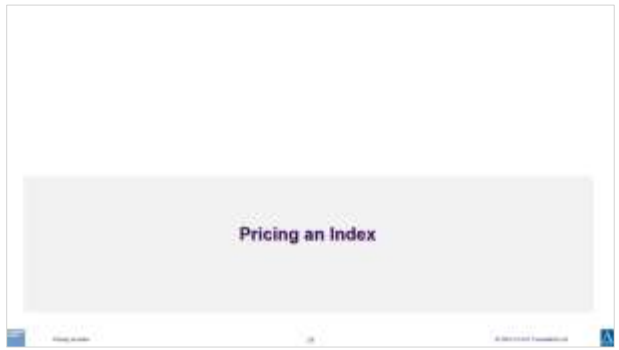
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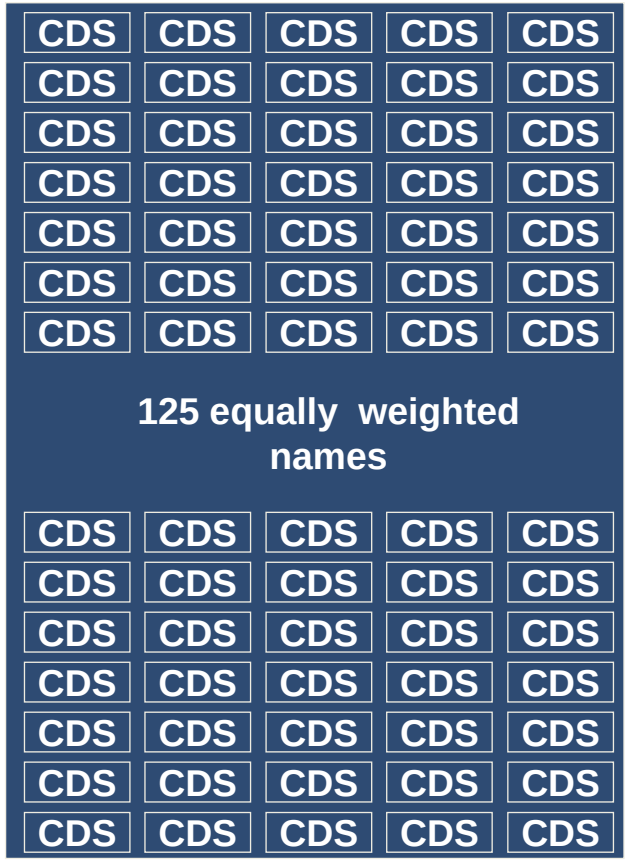
Overview

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Overview of CDS Indices

The big picture:

- A set of market indices for the liquid CDS market
- Analogous to a stock market index:
  - The DJ30, FTSE100, and similar indices provide an indication of where shares are trading
  - The CDX.NA.IG index provides an indication of where CDS are trading
- Indices comprise an equally weighted basket of liquid CDS names
- The index acts in a similar way to a basket of single-name CDS



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CDX and iTraxx Indices

- Original indices were specific to particular dealers e.g. JPMorgan's Hydi (2001)
- CDX and iTraxx were created in 2003/4 to provide a market standard
- Markit Group Ltd now administers both indices since November 2007 when:
  - Markit acquired International Index Company (owners of iTraxx) …
  - …and CDS IndexCo (owners of CDX)

- Licensed End Users (May 2023):
  - Bank of America
  - Barclays Capital
  - BNP Paribas
  - Citibank
  - Citadel Securities
  - Credit Suisse
  - Deutsche Bank
  - Goldman Sachs
  - HSBC
  - JPMorgan
  - Morgan Stanley
  - Nomura
  - RBS
  - UBS
  - Wells Fargo
- CDX (US) and iTraxx (rest of world) provide investors with family of benchmark credit derivative indices…

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Overview of CDX and iTraxx Indices

With liquidity and diversification, the credit derivative indices allow market participants to:

- Gain exposure to various geographies:
  - North America
  - Latin America
  - Europe
  - Asia
  - Japan
  - Australia
  - Emerging Markets
- Gain exposure to specific credit qualities:
  - Investment grade
  - High volatility
  - Crossover
  - High yield
- Go long or short credit
- Gain exposure to specific points on the credit curve:
  - 1 year
  - 3 year
  - 5 year
  - 7 year
  - 10 year
- Gain exposure to specific sectors:
  - Financials
  - Non-Financials
- Invest with and without leverage:
  - Unfunded products
  - Funded products

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CDX Indices

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CDX On-the-Run Main & Sub Baskets

- CDX NA Investment Grade (125 equally-weighted names)
  - CDX NA Investment Grade High Volatility (30 equally-weighted names)
  - CDX NA Investment Grade BBB (90 names in Series 40)
- CDX NA High Yield (100 equally-weighted names)
  - CDX NA High Yield B (36 names in Series 40)
  - CDX NA High Yield BB (58 names in Series 40)
  - CDX NA High Yield Ex-BB (42 names in Series 40)
- CDX Emerging Markets (20 names: all sovereign, weights 2% to 10%)

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CDX.NA.IG Portfolio Characteristics

- 125 most liquid investment-grade credits domiciled in North America
  - Equal weightings (0.8%) for all credits
- Composition of the Index is determined by polling process
  - See next page…
- Diverse pool of credits across industry sectors
  - Represents a broad exposure to the investment grade corporate market
- Static membership
  - Reference Entities cannot be added, and will only be removed upon the triggering of a credit event
- Maturities
  - A new series is launched every six months
    - 20 March 2023 Series 40
    - 20 September 2022 Series 39
    - 21 March 2022 Series 38
    - 20 September 2021 Series 37
    - 22 March 2021 Series 36
    - …
- Investors can buy or sell credit protection

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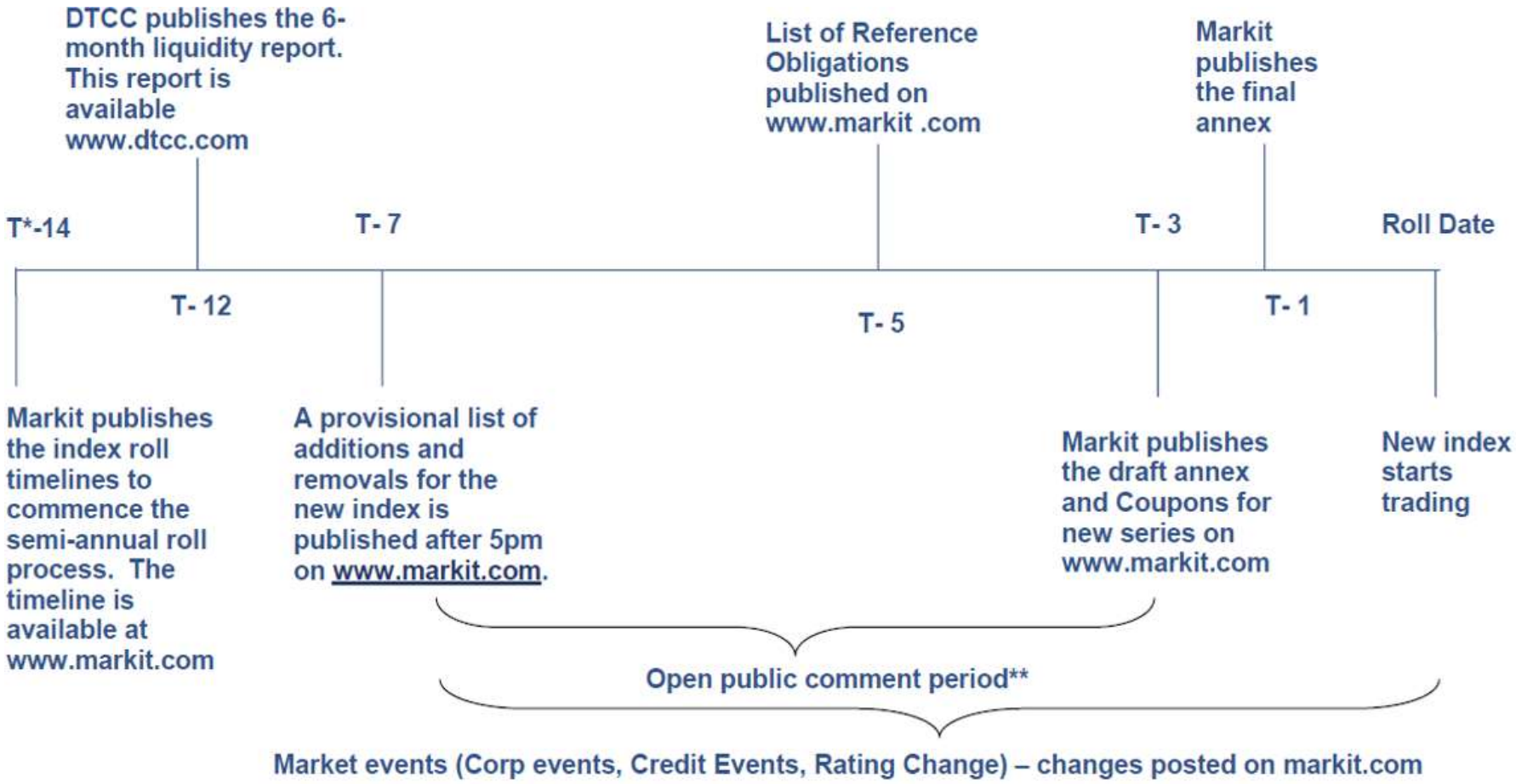
The Polling Process for CDX.NA.IG

- Markit creates a **Liquidity List** using the last six months' DTCC TIW data
  - Entities must have single-name CDS trading under the SNAC conventions
  - The entity or its obligations have an **IG Relevant Rating** that is/are investment grade
- No later than 8 business days prior to the roll date, Markit creates:
  - An exclusion list of any current IG entities that:
    - No longer have an IG Relevant Rating
    - Have a credit event under consideration or in progress
    - Are involved in a merger or other corporate action that renders those entities no longer suitable for inclusion
    - Have single-name CDS contracts ranking in the lowest 30% of the IG Liquidity Rankings
    - Have outstanding debt of less than $100m
    - Are swap dealers
  - A **Liquidity Based Inclusion** list of non-current IG entities that:
    - Are not subject to any of the above exclusion criteria
    - Have single-name CDS contracts ranking in the highest 50% of the IG Liquidity Rankings
    - Have an **Average CDS Spread** over the preceding 90 days less than 5× the current IG index spread
    - Does not have an IG Relevant Rating of BBB- or Baa3 and be on negative watch
    - Favours Rising Stars / Fallen Angels that have just switched
- Markit creates the new IG Index series by:
  - Removing from the list of current IG entities those on the exclusion list
  - Adding sufficient of the most liquid of those entities on the inclusion list to bring the total number to 125
- No later than 7 business days prior to the roll date
  - The provisional composition of the new index is published
  - The list is finalised 3 business days prior to the roll after reviewing any comments received

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Timeline for Index Rolls

Markit CDX.NA.IG, CDX.NA.HY, iTraxx Europe & iTraxx Crossover



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CDX.NA Recent Changes – Series 39 to Series 40

- **08 Mar 2023** – Provisional list of changes:

Removals	Additions
Kohl's Corporation	Radian Group plc

- **Timeline:**

Date*	Local Time	Action
Wed 01. Mar.23		Markit publishes timeline for the index roll process
Wed 08. Mar.23	17:00	A provisional list of additions and removals for the new index will be published on the CDX News site
Fri 10. Mar.23	15:00	The list of reference obligations will be published on the CDX News site
Mon 13. Mar.23	15:00	The composition of the HVOL and other sub-indices will be published on the Markit CDX News site
Mon 13. Mar.23	15:00	Coupons for the IG index and sub-indices will be published on the Markit CDX News site
Tue 14. Mar.23	13:00	Comment period ends for providing feedback about the provisional list of additions and removals, as well as reference obligations
Tue 14. Mar.23	17:00	Draft annexes for the IG index and each sub-index will be published on www.markit.com
Fri 17. Mar.23	17:00	Final annexes for the IG index and each sub-index will be published on www.markit.com
Mon 20. Mar.23		Trading commences

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CDX.NA.IG Series 40 Constituents

Entity	Reference Obligation	Entity	Reference Obligation
Advanced Micro Devices, Inc.	AMD 2.125 01Sep26	CVS Health Corporation	CVSHEA 2.625 15Aug24
Ally Financial Inc.	ALLYFI 5.8 01May25	D.R. Horton, Inc.	DHI 1.4 15Oct27
Altria Group, Inc.	MO 2.625 16Sep26	Danaher Corporation	DHR 3.35 15Sep25
American Electric Power Company, Inc.	AEP 3.2 13Nov27	Darden Restaurants, Inc.	DRI 3.85 01May27
American Express Company	AXP 4.05 03May29	Deere & Company	DE 5.375 16Oct29
American International Group, Inc.	AIG 6.25 01May36 Struc	Dell Inc.	DELLN 7.1 15Apr28
Amgen Inc.	AMGN 3.625 22May24	Devon Energy Corporation	DVN 7.95 15Apr32
Arrow Electronics, Inc.	ARW 7.5 15Jan27	Dominion Energy, Inc.	DOMINEN 4.25 01Jun28
Assured Guaranty Municipal Corp.		DXC Technology Company	DXCTEC 1.8 15Sep26
AT&T Inc.	ATTINC 3.8 15Feb27	EIDP, Inc.	DD 1.7 15Jul25
AutoZone, Inc.	AZO 3.125 21Apr26	Eastman Chemical Company	EMN 7.6 01Feb27
Avnet, Inc.	AVT 3 15May31	Enbridge Inc.	ENB 3.5 10Jun24
Barrick Gold Corporation	ABX 5.8 15Nov34	Energy Transfer LP	ENERTRAE (8) 5.875 15Jan24
Baxter International Inc.	BAX 2.6 15Aug26	Exelon Corporation	EXC 3.4 15Apr26
Berkshire Hathaway Inc.	BRK 3.125 15Mar26	Expedia Group, Inc.	EXPEGRO 6.25 01May25
Best Buy Co., Inc.	BBY 4.45 01Oct28	FedEx Corporation	FDX 3.25 01Apr26
Block Financial LLC	HRB-Fllc 2.5 15Jul28	FirstEnergy Corp.	FE 7.375 15Nov31
BorgWarner Inc.	BWA 3.375 15Mar25	Freeport-McMoRan Inc.	FREEPIN 5 01Sep27
Boston Scientific Corporation	BSX 3.45 01Mar24	General Electric Company	GE 6.75 15Mar32
Bristol-Myers Squibb Company	BMY 6.8 15Nov26	General Mills, Inc.	GIS 4 17Apr25
Campbell Soup Company	CPB 4.15 15Mar28	General Motors Company	GENMC 4.2 01Oct27
Canadian Natural Resources Limited	CNATUR 2.05 15Jul25	Halliburton Company	HAL 2.92 01Mar30
Capital One Bank (USA), National Association	COF-BNKNA 2.28 28Jan26	HCA Inc.	HCA 5.875 15Feb26
Cardinal Health, Inc.	CAH 3.41 15Jun27	Hess Corporation	HESS 3.5 15Jul24
Caterpillar Inc.	CAT 2.6 19Sep29	Honeywell International Inc.	HON 5.7 15Mar36 BondCall
Chubb Limited	CHUBINA 8.875 15Aug29	Host Hotels & Resorts, L.P.	HOSHOT-HSTRES 3.875 01Apr24
Cisco Systems, Inc.	CISCOSY 3.625 04Mar24	HP Inc.	HPINCAA 3 17Jun27
Comcast Corporation	CMCSA 3.7 15Apr24	International Business Machines Corporation	IBM 3.625 12Feb24
Conagra Brands, Inc.	CONABRA 7 01Oct28	International Paper Company	IP 5 15Sep35
ConocoPhillips	COP (3) 5.9 15Oct32	Johnson & Johnson	JNJ 2.9 15Jan28
Cox Communications, Inc.	COX-CommInc 6.8 01Aug28	Johnson Controls International plc	JOHCON 3.625 02Jul24
CSX Corporation	CSX 2.6 01Nov26	Kinder Morgan, Inc.	KINDERM 4.3 01Mar28

Entities that are constituents of the HVOL index are shown in red.

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CDX.NA.IG Series 40 Constituents (cont'd)

Entity	Reference Obligation	Entity	Reference Obligation
Kraft Heinz Foods Company	KRAFHEI 6.375 15Jul28	Sempra Energy	SRE 3.25 15Jun27
Lennar Corporation	LEN 4.875 15Dec23	Simon Property Group, L.P.	SPG-LP 2.65 15Jul30
Lincoln National Corporation	LNC 3.35 09Mar25	Southwest Airlines Co.	LUV 5.125 15Jun27
Lockheed Martin Corporation	LMT 3.55 15Jan26	Targa Resources Partners LP	TGRI-Ptnr (3) 5.5 01Mar30
Loews Corporation	LTR 6 01Feb35	Target Corporation	TGT 2.5 15Apr26
Lowe's Companies, Inc.	LOW 2.5 15Apr26	Teck Resources Limited	TECKRS 6.125 01Oct35 BondCall
M.D.C. Holdings, Inc.	MDC 3.85 15Jan30	The AES Corporation	AES (4) 1.375 15Jan26
Marathon Petroleum Corporation	MARAPET 3.625 15Sep24	The Allstate Corporation	ALL 1.45 15Dec30
Marriott International, Inc.	MAR 3.125 15Jun26	The Boeing Company	BA 2.6 30Oct25
Marsh & Mclennan Companies, Inc.	MMC 5.875 01Aug33	The Dow Chemical Company	DOW 7.375 01Nov29
McDonald's Corporation	MCD 3.5 01Jul27	The Home Depot, Inc.	HD 5.875 16Dec36 BondCall
McKesson Corporation	MCK 7.65 01Mar27	The Kroger Co.	KR 4.5 15Jan29
MetLife, Inc.	MET 3.6 13Nov25	The Procter & Gamble Company	PG 3.25 Mar30
Mondelez International, Inc.	MONDINT 4.125 07May28	The Sherwin-Williams Company	SHW 7.375 01Feb27
Motorola Solutions, Inc.	MOTSOL 7.5 15May25	The Southern Company	SO 3.25 01Jul26
National Rural Utilities Cooperative Finance Corporation	NRUC 8 01Mar32	The Walt Disney Company	THWALT (2) 3.7 15Sep24
Newmont Corporation	NEWMOCO 2.25 01Oct30	The Williams Companies, Inc.	WMB 4.55 24Jun24
NextEra Energy Capital Holdings, Inc.	NXTENG 1.875 15Jan27	T-Mobile USA, Inc.	DT-TMobileUS 4.75 01Feb28
Norfolk Southern Corporation	NSC 3.65 01Aug25	Toll Brothers, Inc.	TOL-FinCorp 4.35 15Feb28
Northrop Grumman Corporation	NORGRM 4.4 01May30	Transcanada Pipelines Limited	TRP 4.875 15Jan26
Omnicom Group Inc.	OMC 3.65 01Nov24	Tyson Foods, Inc.	TSN 3.55 02Jun27
Oracle Corporation	ORCLE 3.25 15Nov27	Union Pacific Corporation	UNP 6.625 01Feb29
Ovintiv Inc.	OVININC 8.125 15Sep30	United Parcel Service, Inc.	UPS-AmericaInc 8.375 01Apr30
Packaging Corporation of America	PACKAM 3 15Dec29	UnitedHealth Group Incorporated	UNH 2.95 15Oct27
Paramount Global	PARAMGL 4.75 15May25	Valero Energy Corporation	VLOC 8.75 15Jun30
Pfizer Inc.	PFE .8 28May25	Verizon Communications Inc.	VRZN 4.125 16Mar27
Prudential Financial, Inc.	PRU 3.878 27Mar28	Walmart Inc.	WALMINC 5.875 05Apr27
PulteGroup, Inc.	PHMG 7.875 15Jun32	Weyerhaeuser Company	WY (5) 7.375 15Mar32
Quest Diagnostics Incorporated	DGX 4.2 30Jun29	Whirlpool Corporation	WHR 4.75 26Feb29
Radian Group Inc.	RDN 4.5 01Oct24		
Raytheon Technologies Corporation	RAYTHTE 3.125 04May27		
Ryder System, Inc.	R 3.875 01Dec23		

Entities that are constituents of the HVOL index are shown in red.

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CDX.NA.HY Series 40 Constituents

Entity	Reference Obligation	Entity	Reference Obligation	Entity	Reference Obligation
American Airlines Group Inc.	AMERAIAI 3.75 01Mar25	Genworth Holdings, Inc.	GENWHOL 6.5 15Jun34	Post Holdings, Inc.	POSTHOL 5.75 01Mar27
American Axle & Manufacturing, Inc.	AXL-Inc 6.25 15Mar26	GFL Environmental Inc.	GFLEN 4.75 15Jun29	R.R. Donnelley & Sons Company	DNY 8.25 01Jul27
Amkor Technology, Inc.	AMKR 6.625 15Sep27	Hilton Domestic Operating Co. Inc.	HILTDOM 5.375 01May25	Rite Aid Corporation	RAD 7.7 15Feb27
Antero Resources Corporation	XTO-ANTERO 5.375 01Mar30	Howmet Aerospace Inc.	HOWMAER 5.125 01Oct24	Royal Caribbean Cruises Ltd.	RCL 3.7 15Mar28
Anywhere Real Estate Group LLC	ANYWHRE (2) 5.25 15Apr30	iHeartCommunications, Inc.	IHEAINC 8.375 01May27	Sabre Holdings Corporation	SABRGLB 9.25 15Apr25
Apache Corporation	APA 4.375 15Oct28	Iron Mountain Incorporated	IRONMOAF 4.875 15Sep27	Safeway Inc.	SWY 7.25 01Feb31
Aramark Services, Inc.	ARAMSER 6.375 01May25	iStar Inc.	ISTAINC 4.25 01Aug25	Sealed Air Corporation	SEE 6.875 15Jul33
Ashland Inc.	ASHLINC 3.375 01Sep31	K. Hovnanian Enterprises, Inc.	HOV-K 7 15Jan19	Service Properties Trust	SERVIPR 5.5 15Dec27
Avient Corporation	AVIECOR (3) 5.75 15May25	KB Home	KBH 6.875 15Jun27	Sirius XM Radio Inc.	SRSRI 5.5 01Jul29
Avis Budget Group, Inc.	CD-AvisBdgt 4.75 01Apr28	Kohl's Corporation	KSS 4.25 17Jul25	Standard Building Solutions Inc.	STANBUI (4) 5 15Feb27
Avon Products, Inc.	AVP 6.95 15Mar43	Lamb Weston Holdings, Inc.	LAMBWES 4.875 15May28	Staples, Inc.	SPLS 10.75 15Apr27
Ball Corporation	BLL 4 15Nov23	Liberty Interactive LLC	LITLC 8.5 15Jul29	TEGNA Inc.	TEGNINC 4.75 15Mar26
Bath & Body Works, Inc.	BATHB 5.25 01Feb28	Louisiana-Pacific Corporation	LPX 3.625 15Mar29	Tenet Healthcare Corporation	THC (2) 6.875 15Nov31
Bausch Health Companies Inc.	BAUSHEA 7 15Jan28	Lumen Technologies, Inc.	LUMETEC (4) 5.625 01Apr25	Teva Pharmaceutical Industries Limited	TEVAPHAG 3.15 01Oct26
Beazer Homes Usa, Inc.	BZH (3) 6.75 15Mar25	Macy's, Inc.	M-RHI 4.5 15Dec34	The ADT Security Corporation	THADT (3) 4.125 15Jun23
Bombardier Inc.	BOMBINC (3) 7.45 01May34	MBIA Inc.	MBI 6.625 01Oct28	The Gap, Inc.	GPS 3.625 01Oct29
Boyd Gaming Corporation	BYD 7.125 01Feb16 BondCall	MGIC Investment Corporation	MGIC 5.25 15Aug28	The Goodyear Tire & Rubber Company	GT 5 31May26
Caesars Entertainment, Inc.	CAESARE 4.625 15Oct29	MGM Resorts International	MGMRI 5.75 15Jun25	The Hertz Corporation	F-Hertz 5 01Dec29
Calpine Corporation	CPN 5.125 15Mar28	MPT Operating Partnership LP	MPW-OPLP 4.625 01Aug29	TransDigm Inc.	TRADIG (2) 6.375 15Jun26
Carnival Corporation	CCL 6.65 15Jan28	Murphy Oil Corporation	MUR 7.05 01May29	Transocean Inc.	RIG 8 01Feb27
Carvana Co.	CARVCO 5.5 15Apr27	Nabors Industries, Inc.	NBR-Inc (2) 5.75 01Feb25	Uber Technologies, Inc.	UBERTEC 7.5 15May25
CCO Holdings, LLC	CHTR-CCOH 5 01Feb28	Navient Corporation	NAVIECO (3) 5.5 15Mar29	Unisys Corporation	UIS (2) 5.5 01Mar21
Chesapeake Energy Corporation	CHK 5.875 01Feb29	Netflix, Inc.	NFLX 4.875 15Jun30	United Airlines Holdings, Inc.	UNITAIBJ 5 01Feb24
Cleveland-Cliffs Inc.	CLEVINC (4) 5.875 01Jun27	New Albertsons L.P.	NEALBE 8 01May31	United Rentals (North America), Inc.	UNITREN 3.875 15Feb31
CommScope, Inc.	CTV 7.125 01Jul28	Newell Brands Inc.	NEWEBRA 4.2 01Apr26	United States Steel Corporation	X 6.65 01Jun37
Community Health Systems, Inc.	COMMHS 6.875 01Apr28	Nordstrom, Inc.	JWN 6.95 15Mar28	Uniti Group Inc.	UNITFIB 4 15Jun24
CSC Holdings, LLC	CVSHLD (3) 5.375 01Feb28	NOVA Chemicals Corporation	NCX 4.875 01Jun24	Universal Health Services, Inc.	UHS (2) 2.65 15Jan32
DaVita Inc.	DAVITIN 4.625 01Jun30	NRG Energy, Inc.	XEL-NRGInc (2) 5.75 15Jan28	Univision Communications Inc.	UVN (2) 8.5 15May21
Delta Air Lines, Inc.	DAL 7.375 15Jan26	Occidental Petroleum Corporation	OXY 5.55 15Mar26	Vistra Operations Company Llc	VISTOPE 5.5 01Sep26
Dish DBS Corporation	DISHN-DDBSC (3) 7.375 01Jul28	Olin Corporation	OLN 5.125 15Sep27	Weatherford International Ltd.	WFT 8.625 30Apr30
Domtar Corporation	DOMC 6.25 01Sep42	OneMain Finance Corporation	ONEMFIAK 6.625 15Jan28	Xerox Corporation	XRX 3.8 15May24
EQM Midstream Partners, LP	EQMMID 4 01Aug24	Pactiv LLC	PACTLLC 7.95 15Dec25	YUM! Brands, Inc.	YUM 3.625 15Mar31
Ford Motor Company	F 4.346 08Dec26	PG&E Corporation	PCG 5 01Jul28		
Frontier Communications Hldgs, LLC	FRCOMM (4) 6.75 01May29	Pitney Bowes Inc.	PBI 4.625 15Mar24		

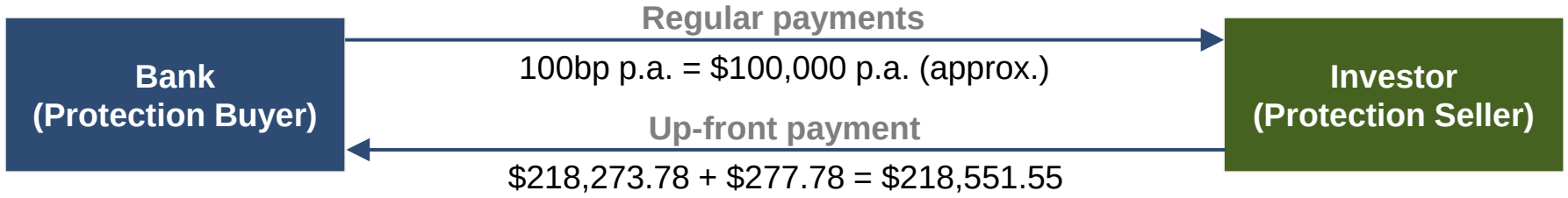
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CDX.NA.IG Series 36 Trading Example

22nd Mar 2021

An investor, believing that credit quality will improve, sells protection on $10m notional of CDX.NA.IG36 Jun 2026 contract.

- Standardised premium: 100.00 bp p.a.
- Quoted premium: 57.20 bp p.a.



- Investor receives 100 bp p.a. for five years to take risk on 125 credits
- Investor up front:
  - **Pays** difference between:
    - 100bp actual income
    - 57.20bp quoted rate
  - **Pays** (Using the ISDA Standard Model): $ 218,273.78
  - **Pays** 1 day of accrued premium: $10m × 100 bp × 1/360: $ 277.78
  - **TOTAL** $ 218,551.55

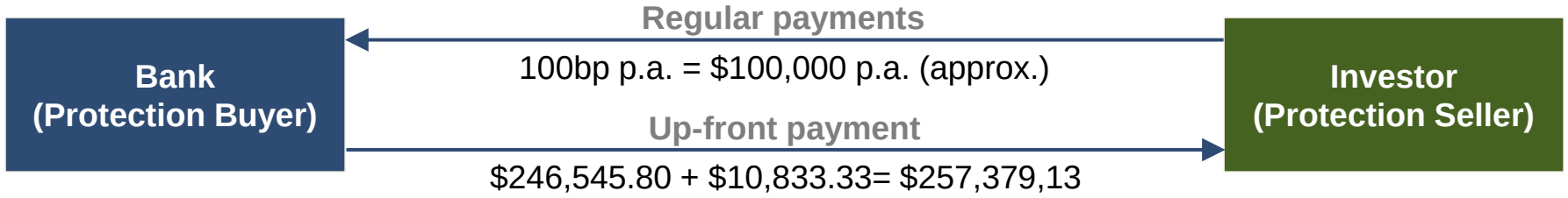
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CDX.NA.IG Series 36 Trading Example

29th Apr 2021

The market has now tightened to 50.87bp, and the investor decides to close out the position by buying back $10m of the CDX.NA.IG36 Jun 2026

- Standardised premium: 100.00 bp p.a.
- Quoted premium: 50.87 bp p.a.



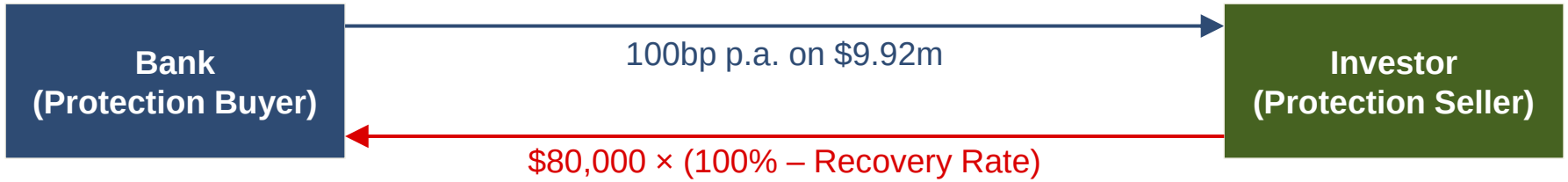
- **Receives** (Using the ISDA Standard Model): $ 246,545.80
- **Receives** 39 days of accrued premium: $10m × 100 bp × 39/360: $ 10,833.33
- **TOTAL** $ 257,379.13

Net profit from the trade: $257,379.13 − $218,551.55 = $38,827.58

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CDX.NA.IG Series 36 Trading Example (cont'd)

- If one name defaults…
  - An auction is held to establish the recovery rate on deliverable bonds
  - Investor pays out pro rata Notional Principal × (100% – Recovery Rate)
  - The affected Reference Entity will be removed from the portfolio…
  - Protection seller continues to earn 100 bp p.a. on a reduced notional



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Re-Versioning After a Credit Event

- When a name is removed from an index series following a credit event, the series continues with one fewer constituent, and the version number is incremented
- The table on the right was published on 22nd March 2023 following the bankruptcy credit event of Diamond Sports Group LLC
- As a result, Diamond was removed from Markit CDX.NA.HY series 34 through 39, and other affected series

Index Name	Series	Version
CDX.NA.HY	34	V11
CDX.NA.HY	35	V3
CDX.NA.HY	36	V3
CDX.NA.HY	37	V3
CDX.NA.HY	38	V3
CDX.NA.HY	39	V2
CDX.NA.HY.B	35	V3
CDX.NA.HY.BB	34	V2
CDX.NA.HY.Ex-BB	35	V3
CDX.NA.HY.Ex-BB	36	V3
CDX.NA.HY.Ex-BB	37	V3
CDX.NA.HY.Ex-BB	38	V3
CDX.NA.HY.Ex-BB	39	V2

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Bloomberg CDSW Page – CDX.NA.IG Illustration



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iTraxx Indices

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iTraxx Europe

- Now administered by Markit
- Licensed End Users (May 2023):
  - Bank of America
  - Barclays Capital
  - BNP Paribas
  - Citigroup
  - Crédit Agricole
  - Credit Suisse
  - Deutsche Bank
  - Goldman Sachs
  - HSBC
  - JPMorgan
  - Morgan Stanley
  - Nomura
  - RBS
  - SocGen
  - UBS
- Static portfolio of 125 equally-weighted CDS on European entities, comprising:
  - 30 Financials
  - 20 Energy
  - 30 Autos & Industrials
  - 25 Consumers
  - 20 TMT
- Indices comprise:
  - iTraxx Europe
    - iTraxx Europe Senior Financials
    - iTraxx Europe Sub Financials
    - iTraxx Europe Non Financials – TMT, Autos & Industrials, Energy, Consumers
  - iTraxx Europe Crossover
- Credit events comprise:
  - Bankruptcy
  - Failure to pay
  - Restructuring
- A new iTraxx MSCI ESG Screened Europe Index was launched in March 2020
  - Comprises constituents of iTraxx but for those excluded for values (e.g. weapons), controversy, and ESG ratings

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iTraxx Europe Series 39

Sector	IHS Markit Ticker	IHS Markit Long Name	ISIN	Maturity	TIER
Autos & Industrials	AIRBSE	AIRBUS SE	XS1050846507	02-Apr-24	SNRFOR
Autos & Industrials	VLVY	AKTIEBOLAGET VOLVO	XS2175848170	26-May-25	SNRFOR
Autos & Industrials	AKZO	AKZO NOBEL N.V.	XS1134519120	07-Nov-24	SNRFOR
Autos & Industrials	ALSTOM	ALSTOM	FR0013453040	14-Oct-26	SNRFOR
Autos & Industrials	AAUK	ANGLO AMERICAN PLC	XS1962513674	11-Mar-26	SNRFOR
Autos & Industrials	ARMLL	ARCELORMITTAL	XS2082324018	19-Nov-25	SNRFOR
Autos & Industrials	AZN	ASTRAZENECA PLC	XS1411404426	12-May-28	SNRFOR
Autos & Industrials	BAPLC	BAE SYSTEMS PLC	USG06940AE30	15-Apr-30	SNRFOR
Autos & Industrials	BASFSE	BASF SE	XS1548422846	11-Mar-25	SNRFOR
Autos & Industrials	BYIF	BAYER AKTIENGESELLSCHAFT	XS2199265617	06-Jul-24	SNRFOR
Autos & Industrials	BMW	BAYERISCHE MOTOREN WERKE AKTIENGESELLSCHAFT	XS1548436473	12-Jul-24	SNRFOR
Autos & Industrials	BOUY	BOUYGUES	FR0013222494	07-Jun-27	SNRFOR
Autos & Industrials	CNHIND	CNH INDUSTRIAL N.V.	US12594KAB89	15-Nov-27	SNRFOR
Autos & Industrials	STGOBN	COMPAGNIE DE SAINT-GOBAIN	XS2517103250	10-Aug-25	SNRFOR
Autos & Industrials	CONTI	CONTINENTAL AKTIENGESELLSCHAFT	XS2056430874	27-Jun-25	SNRFOR
Autos & Industrials	GLCORE	GLENCORE INTERNATIONAL AG	XS1202849086	17-Mar-25	SNRFOR
Autos & Industrials	GSKPLC	GSK PLC	XS0866588527	20-Dec-27	SNRFOR
Autos & Industrials	HEI	HEIDELBERGCEMENT AG	XS1425274484	03-Jun-24	SNRFOR
Autos & Industrials	RWE-Hochtief	HOCHTIEF AKTIENGESELLSCHAFT	DE000A2LQ5M4	03-Jul-25	SNRFOR
Autos & Industrials	HOLCAG	HOLCIM AG	CH0433761274	11-Dec-24	SNRFOR
Autos & Industrials	KDSM	KONINKLIJKE DSM N.V.	XS1041772986	03-Apr-24	SNRFOR
Autos & Industrials	AIRLIQ	L'AIR LIQUIDE SOCIETE ANONYME POUR L'ETUDE ET L'EXPLOITATION DES PROCEDES GEORGES CLAUDE	FR0011439835	06-Sep-23	SNRFOR

… plus other names in the five industry sectors

TMT	VIVESE	VIVENDI SE	FR0013176310	26-May-26	SNRFOR
TMT	VOD	VODAFONE GROUP PUBLIC LIMITED COMPANY	XS1109802568	11-Sep-25	SNRFOR
TMT	WPPGRP-2005	WPP 2005 LIMITED	XS1112013666	22-Sep-26	SNRFOR

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iTraxx Europe Series 39 – Changes

Sector / Index	IHS Markit Ticker	IHS Markit Long Name	IN / OUT
Crossover	AUTIM-ITL	AUTOSTRADE PER L'ITALIA S.P.A.	OUT
Crossover	CMACGM	CMA CGM	OUT
Crossover	ORANSA	ORANO	OUT
Crossover	TDCHOL	TDC HOLDING A/S	OUT
Crossover	BELLACQ	BELLIS ACQUISITION COMPANY PLC	IN
Crossover	FSEKGA	FRESENIUS SE & CO. KGAA	IN
Crossover	PPFTE	PPF TELECOM GROUP B.V.	IN
Crossover	TEOLVOI	TEOLLISUUDEN VOIMA OYJ	IN
TMT	SKYLIM	SKY LIMITED	OUT
Consumers	LODREY	LOUIS DREYFUS COMPANY B.V.	OUT
Consumers	ELTLX	AKTIEBOLAGET ELECTROLUX	IN
Energy	UU	UNITED UTILITIES PLC	IN
MSCI ESG	ELTLX	AKTIEBOLAGET ELECTROLUX	IN
MSCI ESG	UU	UNITED UTILITIES PLC	IN

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iTraxx Europe Crossover Series 39

IHS Markit Ticker	IHS Markit Long Name	ISIN	Maturity	TIER
ACCOR	ACCOR	FR0013399029	04-Feb-26	SNRFOR
ADLEREA	ADLER REAL ESTATE AKTIENGESELLSCHAFT	XS1731858715	06-Feb-24	SNRFOR
AF	AIR FRANCE - KLM	FR0013477254	16-Jan-25	SNRFOR
ALTICFI	ALTICE FINCO S.A.	XS1577952440	15-Jan-28	SECDOM
ALTICEF	ALTICE FRANCE	XS1859337419	01-Feb-27	SECDOM
ARDAPAC	ARDAGH PACKAGING FINANCE PUBLIC LIMITED COMPANY	USG04586AU00	15-Aug-27	SNRFOR
ATSPA	ATLANTIA S.P.A.	XS1558491855	03-Feb-25	SNRFOR
BELLACQ	BELLIS ACQUISITION COMPANY PLC	XS2303071992	16-Feb-26	SECDOM
BOPAFIN	BOPARAN FINANCE PLC	XS1082473395	15-Jul-21	SNRFOR
CWSE	C&W SENIOR FINANCE LIMITED	USG3165UAA90	15-Sep-27	SNRFOR
GROUPE	CASINO GUICHARD-PERRACHON	FR0012074284	05-Aug-26	SNRFOR
CECOAG	CECONOMY AG	XS2356316872	24-Jun-26	SNRFOR
CELLTEL	CELLNEX TELECOM, S.A.	XS1468525057	16-Jan-24	SNRFOR
CIRSFIN	CIRSA FINANCE INTERNATIONAL S.A R.L.	XS1990952779	22-May-25	SECDOM
CLAR	CLARIANT AG	CH0469273541	15-Apr-26	SNRFOR
	… plus further names			
VOLVCAR	VOLVO CAR AB	XS1971935223	02-Apr-24	SNRFOR
ZFEUR	ZF EUROPE FINANCE B.V.	XS2010039977	23-Oct-27	SNRFOR
ZIGGBON	ZIGGO BOND COMPANY B.V.	USN9836ZAA68	28-Feb-30	SECDOM

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iTraxx Asia ex-Japan, Australia, Japan – Series 39

- iTraxx Asia ex-Japan
  - IG index comprises 40 Asian names excluding Japanese entities
  - Maturity: 5yrs

Index	Years	Maturity	Coupon	Recovery Rate (%)
iTraxx Asia ex-Japan	5	20-Jun 2028	100	40

- iTraxx Australia
  - The iTraxx Australia index comprises 25 equally-weighted names on Australian entities
  - Maturity: 5yrs

Index	Years	Maturity	Coupon	Recovery Rate (%)
iTraxx Australia	5	20-Jun 2028	100	40

- iTraxx Japan
  - The iTraxx Japan index comprises 40 equally-weighted names on Japanese entities
  - Maturity: 5yrs

Index	Years	Maturity	Coupon	Recovery Rate (%)
iTraxx Japan	5	20-Jun 2028	100	35

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iTraxx Asia ex-Japan Series 39 – Composition

IHS Markit Ticker	Reference Entity
ALIBGRO	ALIBABA GROUP HOLDING LIMITED
BAIDINC	BAIDU, INC.
BCHINL	BANK OF CHINA LIMITED
BOCAV	BOC AVIATION LIMITED
CHINCIAA	CHINA CINDA (HK) HOLDINGS COMPANY LIMITED
CCBC	CHINA CONSTRUCTION BANK CORPORATION
CHINDEAC	CHINA DEVELOPMENT BANK
CHUARO	CHINA HUARONG INTERNATIONAL HOLDINGS LIMITED
CORP3177	CHINA NATIONAL CHEMICAL CORPORATION LIMITED
CHINORI	CHINA ORIENT ASSET MANAGEMENT (INTERNATIONAL) HOLDING LIMITED
CHINAPE	CHINA PETROCHEMICAL CORPORATION
CITILIM	CITIC LIMITED
DBSSP-DBSBank	DBS BANK LTD.
EXIM	EXPORT-IMPORT BANK OF INDIA
HUWHY	HUTCHISON WHAMPOA LIMITED
ICBCHN	INDUSTRIAL AND COMMERCIAL BANK OF CHINA LIMITED
THAI	KINGDOM OF THAILAND
KOREAT	KT CORPORATION
MALAYS	MALAYSIA
CHINA	PEOPLE'S REPUBLIC OF CHINA

IHS Markit Ticker	Reference Entity
PLNIJ	PERUSAHAAN PERSEROAN (PERSERO) PT PERUSAHAAN LISTRIK NEGARA
PETROL	PETROLIAM NASIONAL BERHAD (PETRONAS)
POSCHOL	POSCO HOLDINGS INC.
PTINDAJ	PT INDONESIA ASAHAN ALUMINIUM PERSERO
PTPER	PT PERTAMINA (PERSERO)
RELIND	RELIANCE INDUSTRIES LIMITED
INDON	REPUBLIC OF INDONESIA
KOREA	REPUBLIC OF KOREA
PHILIP	REPUBLIC OF THE PHILIPPINES
STSP	SINGAPORE TELECOMMUNICATIONS LIMITED
SBIIN-StateBkIn	STATE BANK OF INDIA
SGCC	STATE GRID CORPORATION OF CHINA
SUNHUN-Prop	SUN HUNG KAI PROPERTIES LIMITED
SWIRE	SWIRE PACIFIC LIMITED
TENCHOL	TENCENT HOLDINGS LIMITED
EIBC	THE EXPORT-IMPORT BANK OF CHINA
EIBKOR	THE EXPORT-IMPORT BANK OF KOREA
HKLAND-Co	THE HONGKONG LAND COMPANY, LIMITED
VANKREA	VANKE REAL ESTATE (HONG KONG) COMPANY LIMITED
CMBKKO	WOORI BANK

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iTraxx Japan Series 39 – Composition

IHS Markit Ticker	Reference Entity
ACOM	ACOM CO., LTD.
JUSCO	AEON CO., LTD.
ANAHOL	ANA HOLDINGS INC.
FUJITS	FUJITSU LIMITED
HITACH	HITACHI, LTD.
HNDA	HONDA MOTOR CO., LTD.
CITOH	ITOCHU CORPORATION
JAPTOB	JAPAN TOBACCO INC.
JFEHLD	JFE HOLDINGS, INC.
KAWHI	KAWASAKI HEAVY INDUSTRIES, LTD.
KAWKIS	KAWASAKI KISEN KAISHA, LTD.
KINTGRO	KINTETSU GROUP HOLDINGS CO., LTD.
KOBSTL	KOBE STEEL, LTD.
KOMATS	KOMATSU LTD.
MARUB	MARUBENI CORPORATION
MITCO	MITSUBISHI CORPORATION
MITHI	MITSUBISHI HEAVY INDUSTRIES, LTD.
MITSCO	MITSUI & CO., LTD.
MITTOA	MITSUI CHEMICALS, INC.
MITSOL	MITSUI O.S.K. LINES, LTD.

IHS Markit Ticker	Reference Entity
NECORP	NEC CORPORATION
NPG-NPI	NIPPON PAPER INDUSTRIES CO.,LTD.
NIPPSTAA	NIPPON STEEL CORPORATION
NIPYU	NIPPON YUSEN KABUSHIKI KAISHA
NSANY	NISSAN MOTOR CO., LTD.
OJIHOL	OJI HOLDINGS CORPORATION
ORIX	ORIX CORPORATION
PANASHO	PANASONIC HOLDINGS CORPORATION
RAKUGRO	RAKUTEN GROUP, INC.
RICOH	RICOH COMPANY, LTD.
SHIMIZ	SHIMIZU CORPORATION
SOFTGRO	SOFTBANK GROUP CORP.
SONYGRO	SONY GROUP CORPORATION
SUMICH	SUMITOMO CHEMICAL COMPANY, LIMITED
SUMI	SUMITOMO CORPORATION
SUMIRD	SUMITOMO REALTY & DEVELOPMENT CO., LTD.
TFARMA	TAKEDA PHARMACEUTICAL COMPANY LIMITED
TOKYOEL	TOKYO ELECTRIC POWER COMPANY HOLDINGS, INCORPORATED
TOSH	TOSHIBA CORPORATION
TOYOTA	TOYOTA MOTOR CORPORATION

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iTraxx Australia Series 39 – Composition

IHS Markit Ticker	Reference Entity
APINFR	APA INFRASTRUCTURE LIMITED
AURINET	AURIZON NETWORK PTY LTD
AUSGFIN	AUSGRID FINANCE PTY LTD
AUSNSER	AUSNET SERVICES HOLDINGS PTY LTD
ANZ	AUSTRALIA AND NEW ZEALAND BANKING GROUP LIMITED
BHPGR	BHP GROUP LIMITED
CBA	COMMONWEALTH BANK OF AUSTRALIA
GOODMAF	GOODMAN FUNDS MANAGEMENT LIMITED AS RESPONSIBLE ENTITY OF THE GOODMAN INDUSTRIAL TRUST
MQB	MACQUARIE BANK LIMITED
NAB	NATIONAL AUSTRALIA BANK LIMITED
NCM	NEWCREST MINING LIMITED
QANTAS	QANTAS AIRWAYS LIMITED
RIOLN-Ltd	RIO TINTO LIMITED
STOAU	SANTOS LIMITED
SCENTMA	SCENTRE MANAGEMENT LIMITED AS RESPONSIBLE ENTITY OF SCENTRE GROUP TRUST 1
SGSPAUS	SGSP (AUSTRALIA) ASSETS PTY LTD
STSP-OptusPty	SINGTEL OPTUS PTY LIMITED
SOCAC	SOUTHERN CROSS AIRPORTS CORPORATION HOLDINGS LIMITED
TELECO	TELSTRA CORPORATION LIMITED
TCLAU-FIN	TRANSURBAN FINANCE COMPANY PTY LTD
VCENTR	VICINITY CENTRES RE LTD AS RESPONSIBLE ENTITY OF VICINITY CENTRES TRUST
WESAU	WESFARMERS LIMITED
WSTP	WESTPAC BANKING CORPORATION
WOODSEN	WOODSIDE ENERGY GROUP LTD.
WOOLGRO	WOOLWORTHS GROUP LIMITED

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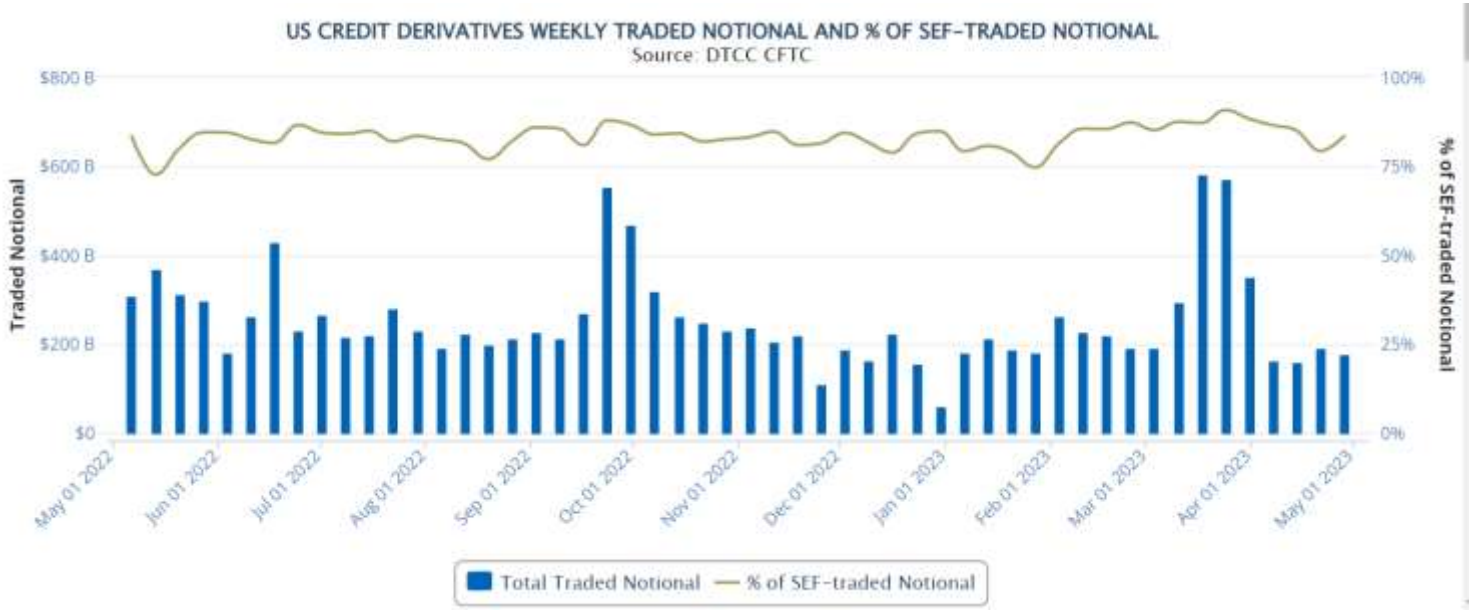
CDX and iTraxx SEFs

The following SEFs provide swap execution facilities:

- Bloomberg
- GFI
- ICAP
- ICE Swap Trade
- Javelin
- MarketAxess
- Tera Exchange
- Tradeweb
- Tradition
- Tullett

CDS Index Notional Volume and Trade Count

YTD Ending	April 28, 2023		April 29, 2022	
	Traded Notional (US$ trillions)	Trade Count (thousands)	Traded Notional (US$ trillions)	Trade Count (thousands)
Credit Derivatives Total	4.35	129.2	5.56	143.1
CDX HY	0.72	36.4	0.84	39.1
CDX IG	1.56	31.5	1.89	33.4
iTraxx Europe	1.03	23.4	1.40	25.3
Other	1.04	37.9	1.43	45.3
Cleared	3.79	116.5	4.61	124.4
ON SEF	3.70	114.3	4.56	123.4



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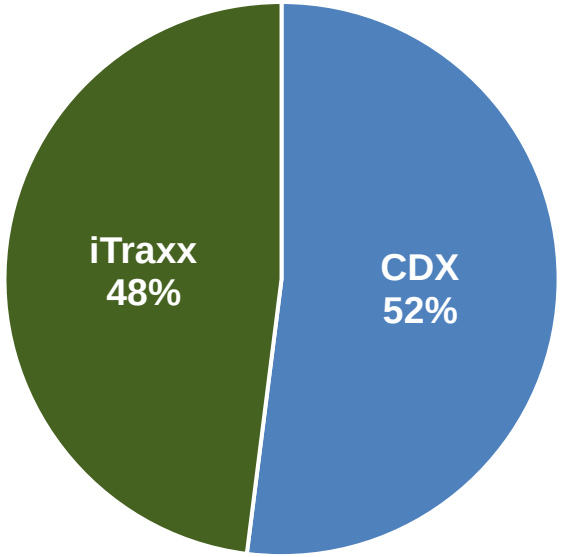
Market Statistics

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Market Statistics

Index/Index Tranche	Average Daily Notional ($m)	
	Untranched	Tranched
CDX.NA.IG	46,700	275
ITRAXX EUROPE	37,825	600
CDX.NA.HY	10,975	175
ITRAXX EUROPE CROSSOVER	9,650	75
ITRAXX EUROPE SENIOR FINANCIALS	5,550	
CDX.EM	1,925	
ITRAXX EUROPE SUB FINANCIALS	900	
ITRAXX ASIA EX-JAPAN	350	
ITRAXX JAPAN	300	
ITRAXX AUSTRALIA	175	
IBOXX USD LIQUID HIGH YIELD	150	
CMBX	50	
TOTAL	**114,550**	**1,125**

Analysis by Index Group



Source: DTCC
27 August 2022 through 24 February 2023

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Pricing an Index

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Pricing an Index – 1

- From the section on pricing a single-name CDS, the *N*-year premium $s_{i,N}$ for a single-name CDS on entity *i* is:

$$s_{i,N} \times \sum_{j=1}^{N} T_{j-1,j} v_j PS_{i,j} = (1-R) \sum_{j=1}^{N} v_j PD_{i,j}$$

- From the same section, we also have the formula for the *PV01* (on a notional of $1):

$$PV01_i = 0.0001 \times \sum_{j=1}^{N} T_{j-1,j} v_j PS_{i,j}$$

- Combining these two formulas, we get:

$$s_{i,N} \times PV01_i = 0.0001 \times (1-R) \sum_{j=1}^{N} v_j PD_{i,j}$$

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Pricing an Index – 2

- Now let's extend the first formula on the previous page in order to price an *N*-year index I_N:

$$I_N \times \sum_{i=1}^{M}\sum_{j=1}^{N} T_{j-1,j} v_j PS_{i,j} = (1-R)\sum_{i=1}^{M}\sum_{j=1}^{N} v_j PD_{i,j}$$

$$\therefore I_N = \frac{(1-R)\sum_{i=1}^{M}\sum_{j=1}^{N} v_j PD_{i,j}}{\sum_{i=1}^{M}\sum_{j=1}^{N} T_{j-1,j} v_j PS_{i,j}}$$

- From the previous page:

$$\sum_{i=1}^{M} s_{i,N} \times PV01_i = 0.0001 \times (1-R)\sum_{i=1}^{M}\sum_{j=1}^{N} v_j PD_{i,j}$$

- … and also:

$$\sum_{i=1}^{M} PV01_i = 0.0001 \times \sum_{i=1}^{M}\sum_{j=1}^{N} T_{j-1,j} v_j PS_{i,j}$$

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Pricing an Index – 3

- Substituting the last two formulas into the expression for IN gives:

$$\therefore I_N = \frac{\sum_{i=1}^{M} s_{i,N} \times PV01_i}{\sum_{i=1}^{M} PV01_i}$$

- This means that:

Index Premium = Weighted Average of the Constituent Single-Name Premiums

… where the weights are the PV01s of the constituent CDSs

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